bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Vocal efficiency in crows

Claudia A.F. Wascher[1]* & Mason Youngblood

[1]Behavioural Ecology Research Group, School of Life Sciences, Anglia Ruskin University, United Kingdom
[2] Institute for Advanced Computational Science, Stony Brook University, United States of America

*Corresponding author: Claudia A.F. Wascher: School of Life Sciences, Anglia Ruskin University, East Road, Cambridge, CB1 1PT United Kingdom; Phone: +44 (0) 1223 698270;
e-mail: claudia.wascher@gmail.com

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Abstract

Many communicative systems have been selected for efficiency, shaped by the trade-off between information transmission and energetic or temporal constraints. Linguistic laws such as Menzerath's law—predicting shorter elements in longer sequences—have emerged as widespread principles across vocal communication in many species. While these laws have been predominantly studied at the species level, the influence of individual and social factors remain underexplored. In this study, we investigated adherence to Menzerath's law in the vocal communication of carrion crows, *Corvus corone corone*, hooded crows, *Corvus corone cornix* and hybrids. Our findings show that crow call sequences adhere to Menzerath's law, with shorter calls occurring in longer sequences, demonstrating structural efficiency in vocal communication. In carrion crows specifically, we analysed call sequences in relation to individual characteristics (sex, age) and social variables (group size, dominance status, strength of affiliative relationships). Interestingly, adherence to Menzerath's law was stronger in males and younger individuals, while no effects were found for group size, dominance, or affiliative relationships. This study provides the first evidence of Menzerath's law in corvid vocal communication and suggests that individual-level traits, rather than broader social dynamics, may shape vocal efficiency. These findings broaden our understanding of widespread principles in animal communication and raise new questions about the ontogeny and flexibility of vocal efficiency in complex social species.

Key words: carrion crows, communicative efficiency, hooded crows, hybrids, linguistic laws, vocal communication

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Introduction

Biological systems have evolved to operate efficiently under the constraints of energy and time, ensuring survival and reproductive success (Igamberdiev, 2023). This is also the case for a wide range of communication systems, such as human language (Gibson et al., 2019) and vocal communication in non-human animals (Endler, 1993; Semple et al., 2022). Communication systems typically evolve to convey critical information accurately, which often requires an increase of signal complexity. For example in the context of mating, increased signal complexity is associated with increased mating success (Choi et al., 2022; Pollard & Blumstein, 2012). In an anti-predator context, complex signals can convey information such as predator type and level of risk (Suzuki, 2014) and social factors such as increased group sizes, and cohesiveness of social bonds requires increasingly complex communication systems such as increases in vocal repertoires or diversity of signals (Peckre et al., 2019). The need for complexity for information transmission needs to be balanced with time and energy constraints, consequently selecting communication systems for efficiency and allowing to optimize the benefits resulting from information transmission while reducing the costs of signal production (Endler, 1993).

One widespread way to reduce costs of communicating an increase signalling efficiency is the reduction of communication time, for example by minimizing the length of a signal. In the context of human language, this efficiency is traditionally quantified as linguistic laws, such as Zipf's law of brevity, which predicts a negative relationship between the length of words and how often they are used (Zipf, 1949) and Menzerath's law, which predicts that longer sequences will be comprised of shorter items to balance production costs (Menzerath, 1954). In recent years, empirical evidence is accumulating suggesting that these information theoretical principles are widespread across a wide range of taxa and communication modalities. Menzerath's law is common for example in bird song (Favaro et al., 2020; James et al., 2021; Lewis et al., 2023; Youngblood, 2024), primates (Gustison et al., 2016; Huang et al., 2020; Valente et al., 2021), bats (Zhang et al., 2024), or whale vocalisations (Youngblood, 2025). Adherence to Menzerath's law is not limited to vocal communication but can also be seen in gestural communication (Heesen et al., 2019; Safryghin et al., 2022).

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

While adherence to linguistic laws seem widespread in non-human animal vocal communication systems, studying social and ecological factors shaping the phenomenon is of interest to further understand the evolutionary pressures shaping efficiency in animal communication systems. Marmoset monkeys, *Callithrix jacchus*, can be trained in a vocal conditioning experiment adopt several vocal compression strategies such as increasing call rate, decreasing call duration and increasing the proportion of short calls, which are consistent with the principles of Zipf's law of brevity (Risueno-Segovia et al., 2023). However, previous studies on common marmoset and the golden-backed uakari, *Cacajao melanocephalus*, fail to find empirical support for Zipf's law in vocalisations (Bezerra et al., 2011). In rock hyraxes, *Procavia capensis*, Zipf's law could only be shown in male but not female calls (Demartsev et al., 2019). In different species Menzerath's Law is only present in certain call-types (tarsiers monkeys, *Tarsius spectrumgurskyae*, titi monkeys, *Plecturocebus cupreus* and gibbons, *Plecturocebus cupreus:* Clink & Lau, 2020; Hainan frilled treefrogs, *Kurixalus hainanus:* Deng et al., 2024).

In the present study we investigated adherence to Menzerath's law in two subspecies of carrion crows, carrion crows, *Corvus corone corone*, hooded crows, *Corvis corone cornix* and hybrids (thereafter crows). Crows belong to the family of corvids, a large family of birds belonging to suborder of oscine passerine birds, with vocal communication playing an important role in the socio-ecology of corvids (Wascher & Reynolds, 2025). Corvids mostly produce calls, i.e., short, distinct vocalisations and not song, i.e., heterogeneous, combinatory vocalisations consisting of notes or phrases that are arranged in a specific order and often repeated (Sandoval & Graham, 2025). Previously, Menzerath's law was mostly investigated in song, not calls. Further, Menzerath's law is mostly investigated on species level and whether it is present in a certain species or not and only recently individual level effects have been described in Java sparrows, *Padda oryzivora* (Lewis et al., 2023). In the present study, we are aiming to investigate how individual factors, such as sex and age as well as social factors like group size, dominance and affiliative relationship status affect adherence to Menzerath's law. Corvids show high

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

113 levels of flexibility in their vocalisations, for example common ravens, *Corvus corax*, flexibly adjust calls
114 to audience composition (Szipl et al., 2018) and show similarity in long-distance calls between pair-
115 partners (Luef et al., 2017). Carrion crows have volitional control over calls (Brecht et al., 2019) and can
116 flexibly produce a variable number of vocalisations, with the acoustic features of the first vocalisation
117 being predictive of the total number of calls in a sequence (Liao et al., 2024). Corvids are open-end
118 vocal learners, which means they can acquire new vocalizations throughout their lifetime and not only
119 in a specific sensitive period (Brenowitz et al., 1997). Many corvid species are mimics, allowing them to
120 copy sounds from the environment or conspecifics (Wascher et al., 2025). As vocal communication
121 plays an important role in allowing corvids to navigate social and ecological challenges, we expect
122 widespread rules of signalling efficiency to apply to their calls. We focus on the Menzerath-Altmann
123 law—a precise and more robust mathematical form of Menzerath's law (Altmann, 1980; Torre et al.,
124 2021). Data was collected from carrion and hooded crows as well as hybrids, although we mostly focus
125 on carrion crows as sample size for hooded crows and hybrids is very small (two individuals each).
126
127 In carrion crows, we further investigate whether within the species adherence to Menzerath's law
128 varies, depending on individual factors, such as sex and age and social factors (group size, dominance
129 status and strength of affiliative relationships). We expect weaker adherence to Menzerath's law in
130 younger individuals compared to adults, as they are expected to still be learning vocalisations patterns.
131 As vocal communication is important for both male and female crows, we do not have specific
132 predictions regarding a sex-based difference how strongly Menzerath's law is observed in carrion
133 crows. Group size is often considered a driving factor of 'social complexity' and associated with this
134 'vocal complexity' (Freeberg & Krams, 2015; Krams et al., 2012; Manser et al., 2014). We do expect a
135 higher need for efficiency in larger groups to allow for coordinated vocalisations and behaviour, hence
136 adherence to Menzerath's law is expected to be higher in larger groups. Lastly, we also investigate
137 effects of dominance and strength of affiliative relationships. We expect the ability to communicate
138 efficiently and hence adherence to Menzerath's law to be positively correlated with dominance status
139 and strength of affiliative relationships in the group.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Methods

Ethical note

The current study was conducted in two populations of captive crows, housed in large outdoor enclosures in Navafría, Castilla y León, Spain (42°36'33 N 5°26'56) and Grünau, Upper Austria (47°51'02 N 13°57'44 E). Crows remained in captivity prior to, during and remained after the presented study. Aviary sizes varied between populations and groups from 20 to 72 m^2 and are always equipped with wooden perches, natural vegetation and rocks. An enriched diet consisting of fruit, vegetables, bread, meat and milk products is provided daily. In both locations water is available *ad libitum* for both drinking and bathing. Keeping of captive birds in Spain was authorized by Junta de Castilla y León (núcleo zoologicó 005074) and in Austria under a licence issued to the Cumberland game park Grünau (AT00009917).

Study subjects

Data for the present study has been collected from 2010 to 2015. As data for the present study has been taken over the extended period time, group composition and study subjects changed in different phases, which is summed up in table 1. Most study subjects are carrion crows, however in the Austrian population there were also two hooded crows and two hybrids (table 1). Since a recent taxonomic update, carrion crows and hooded crows are considered subspecies of the same species (AviList Core Team, 2025). Birds were kept in different group compositions, which reflect natural conditions in the wild: (A) 'flock' or 'family': groups of three and more individuals, mostly juvenile individuals grouped together but in one case a pair successfully reproducing; (B) 'pair': Adult individuals are mostly kept in male-female pairs; in some cases also in trios and due to death of partner for a limited amount of time as singles. All birds were hatched in the wild and most were hand-raised and brought into captivity before fledging, except Valencia and Xufa, who hatched in captivity. Groups were visually but not acoustically separated from each other. For more information on study subjects a see Wascher (2021) and Wascher et al. (2019). As an estimate for strength of affiliative relationships for each subject we calculated a Composite Sociality Index (CSI) and as an estimate for dominance rank, we calculated an

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Elo-rating (ELO) for each subject. We recorded a total of 2,079 individual focal observations (899 in the Austrian population; 1,180 in the Spanish population) lasting five minutes each. All occurring behaviours were recorded but for this study, we focused on the frequencies of agonistic behaviour (threat, chase flight and fight) and affiliative behaviours (allopreen and contact sit). We recorded the identity, role (initiator/receiver) of interacting individuals and the outcome of the agonistic interaction (winner/loser), with the loser of an agonistic interaction defined as the individual that retreated. CSI was calculated according to (Silk et al., 2010). Following Archie et al. (2014), we corrected for a different observational effort between individuals by regressing CSI against the number of observations per individual. We calculated a mean CSI for each individual across all possible dyads, representing the average strength of affiliative relationships. The relative success levels of individuals in agonistic encounters was calculated as an Elo-rating in the R package 'aniDom' (version 0.1.4; Sánchez-Tójar et al., 2018). Each individual was rated based on the outcome of each discrete interaction (winner/loser) and the (predicted) probability of that outcome occurring (Neumann et al., 2011). Similar to above, we corrected for a different observational effort between individuals by regressing Elo-rating against the number of observations per individual. Details of behavioural data collection and calculation of CSI and Elo-rating are described in Wascher (2021) and Wascher et al. (2019).

Recording vocalisations and data analysis

Vocalisations have been recorded during regular behavioural observations using either a Sennheiser ME67 directional microphone on a K6 module, and a 13 Marantz PMD661 digital recorder or a SM1 Wildlife acoustics SongMeter. Recordings were opportunistically taken when an observer was present in the enclosure and hence recording durations differ between individuals (Table 1). Crows were generally habituated to the human observer. The human observer identified the calling individual by speaking on the recording.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

194 **Table 1:** Study subjects. Year of hatching, sex, species, population and group compositions of all

195 individuals contributing to the present study. Dates refer to the times recordings were made in a

196 specific group composition and the number of calls and number of sequences recorded.

Individual	Age	Sex	ELOrating	CSI	Subspecies	Population	Group	Dates
			160.476	0.865				2013-01-06
Alicante	2010	F			Carrion	Spain	HL4	2013-11-15
			-254.001	3.764			Baerchen,	2010-04-11
Baerchen	2008	M			Carrion	Austria	Peter	2015-07-04
			-109.88	3.093				2013-01-11
Bluepoint	2012	M			Carrion	Spain	VR	2013-09-25
			324.532	2.431				2013-01-12
Bluestripes	2012	M			Carrion	Spain	VR	2013-06-16
			63.983	2.415				2012-12-27
Cabezon	2010	M			Carrion	Spain	VL	2014-04-19
			-582.255	3.218			Franz,	2010-04-22
Franz	2007	M			Carrion	Austria	Toeffel	2012-07-26
			116.241	0.477			Gabi, Klaus,	2010-04-09
Gabi	2007	F			Carrion	Austria	Hugo, Resa	2010-09-10
							Gabi,	2011-09-10
							Klaus,Resa	2012-09-10
								2012-11-10
							Gabi	2012-11-15
			-216.389	2.181			Gertrude,	02.10.2011
Gertrude	2011	F			Hooded	Austria	Walter	01.06.2013
			-355.748	0.636				2012-12-30
GreenO	2011	M			Carrion	Spain	HL4	2013-04-26
								2013-05-04
							HL3	2013-09-24
							GreenO,	2013-11-12
							Whitestripes	2014-04-15
			-1.455	0.706				2013-01-06
GreenZ	2011	M			Carrion	Spain	HL4	2014-04-19
								2013-05-11
							HL3	2013-11-15
							GreenZ,	2014-01-31
							White	2014-04-19
			-156.827	4.486			Gabi, Klaus,	2010-04-09
Hugo	Unknown*	M			Carrion	Austria	Hugo, Resa	2010-09-08
			-53.24	1.654				2012-12-31
Ibiza	2010	F			Carrion	Spain	HR	2013-05-09
								2013-07-01
							HR4	2014-12-25
			371.871	0.368			Gabi, Klaus,	2011-09-07
Klaus	2009	M			Carrion	Austria	Resa	2012-01-10
								2012-01-14
							Gabi, Klaus	2012-09-21
			46.029	3.29				2012-12-31
Martinez	2010	M			Carrion	Spain	HR	2013-06-16
							HR4	2013-06-25

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

								2014-12-25
			-139.34	0.107			Munin,	2015-05-28
Munin	2014	F			Hybrid	Austria	Momo, Willi	2015-05-30
								2012-07-29
Nino	Unknown*	M	0	0	Hooded	Austria	Nino	2012-07-29
							Olaf,	2010-04-12
Olaf	2005	F	0	0	Hybrid	Austria	Ruediger	2010-05-01
			-266.528	4.636			Baerchen,	2010-05-13
Peter	2007	F			Carrion	Austria	Peter	2015-07-01
			-108	3.706				2013-01-15
Pobla	2010	F			Carrion	Spain	VL	2014-04-19
			-166.673	2.63				2013-01-11
Redcross	2012	F			Carrion	Spain	VR	2013-09-15
			77.096	3.331				2013-01-06
Redpoint	2012	M			Carrion	Spain	VR	2013-09-25
			43.234	2.431				2013-01-06
Redstripes	2012	M			Carrion	Spain	VR	2013-01-06
			80.771				Gabi, Klaus,	2012-01-14
Resa	2009	F		0	Carrion	Austria	Hugo, Resa	2012-07-17
							Olaf,	2010-04-28
Ruediger	Unknown*	M	0	0	Carrion	Austria	Ruediger	2012-07-12
			-72.342	9.957			Franz,	2010-04-22
Toeffel	2008	F			Carrion	Austria	Toeffel	2012-11-11
			23.732	0.406				2013-06-25
Valencia	2013	M			Carrion	Spain	HR4	2014-12-25
			216.389	1.144			Gertrude,	2011-09-07
Walter	2011	M			Carrion	Austria	Walter	2013-11-27
							Baerbel,	2015-05-27
							Walter	2015-07-05
			241.758	1.189				2012-12-30
White	2008	F			Carrion	Spain	HL4	2013-03-28
								2013-05-04
							HL3	2013-11-15
							GreenZ,	2014-01-31
							White	2014-04-16
			-46.987	2.344				2013-01-23
Whitecross	2012	M			Carrion	Spain	VR	2013-09-21
								2014-04-12
							Wild	2014-04-18
			58.623	1.042				2013-09-10
Whitepoint	2012	M			Carrion	Spain	VR	2013-09-21
			-156.653	0.897				2013-01-11
Whitestripes	2012	F			Carrion	Spain	VR	2014-04-15
			139.34	0.107			Munin,	2015-07-04
Willi	2012	M			Carrion	Austria	Momo, Willi	2015-07-04
								2013-11-26
							Toeffel, Willi	2013-11-27
			-102.11	1.08				2013-06-25
Xufa	2013	F			Carrion	Spain	HR4	2013-06-25

*For individuals with unknown age, we have used the mean ages of all the other birds.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Onsets and offsets of individual calls have been annotated by CAFW using Audacity® or Raven Pro (K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology, 2024). Vocalizations from unknown callers, overlapping vocalizations, incomplete sequences, and single call sequences have been excluded for the present study. Call sequences can be intuitively recognized by human observers, however for the present study we have classified calls with more than one second inter-call duration as separate sequences. This was manually validated by CAFW. In total, we recorded 12,123 calls in 3,726 complete sequences. For details about calls and sequences per individual, see Table 1.

Statistical analysis

All models were fit using the lme4 (v1.1-35.1) (Bates et al., 2015) package in R (v4.3.1) (R Core Team, 2019). To avoid the many problems associated with p-values, we report mean estimates and 95% Wald confidence intervals and interpret intervals that do not overlap zero as indicating a strong effect. All reported models were manually checked for convergence.

We focus on the Menzerath-Altmann law—a precise and more robust mathematical form of Menzerath's law (Altmann, 1980; Torre et al., 2021). Here is the standard form of the Menzerath-Altmann law where y is the duration of elements within a sequence composed of x elements, and a, b, and c are parameters controlling the shape of the relationship.

(Equation 1)

$$y = ax^{b}e^{cx}$$

If we set c to 0 and apply some simple algebra, we get a linear model that is the most common form of the law in contemporary linguistics (Hou et al., 2017).

(Equation 2)

$$\ln(y) = \ln(a) + b*\ln(x)$$

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

We will follow other studies of the Menzerath-Altmann law in nonhuman animals and use the above linear model (Clink & Lau, 2020; Favaro et al., 2020; Gustison et al., 2016, p. 201; James et al., 2021; Lewis et al., 2023; Stepanov et al., 2023; Vradi, 2021; Youngblood, 2024, 2025). y is usually the mean duration of elements within sequences, but we will use the full distribution of call durations within sequences (Youngblood, 2024; Youngblood; 2025) to avoid spurious "regression to the mean" effects (Ferrer-i-Cancho et al., 2013; Gustison et al., 2016; Milička, 2023) and better capture uncertainty in the models (Youngblood, 2024). We also follow other work in excluding single-element sequences (i.e., with a length of one) from the analysis, which have been shown to depart from Menzerath's law (G. Torre et al., 2021; Heesen et al., 2019; Hernández-Fernández et al., 2019; Torre et al., 2019).

The "simple" model, used to assess Menzerath's law separately in carrion crows, hooded crows, and hybrids of carrion and hooded crows, had the following specification:

(Equation 3)

$$\ln(\text{call_duration}) \sim \ln(\text{sequence_length}) + (1|\text{individual/sequence})$$

Nested varying intercepts for sequences and individuals are included to account for the repeated measurement of call durations within sequences and for individual variation in call durations. It is also worth noting that we applied the simple model separately to each species, rather than analysing them together with species as a fixed effect to assess species differences. In this case, given the dramatic disparity in sample sizes between the three species, we think it is unwise to make conclusions about the relative strength of Menzerath's law, and focus on the presence of it instead.

The "complex" model, used to assess individual variation in Menzerath's law based on demographic factors, had three potential specifications:

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

(Equation 4)

ln(call_duration) ~ ln(sequence_length) + sex + group_size + age + elo + csi +

(1|population/group/individual/sequence)

(Equation 5)

ln(call_duration) ~ ln(sequence_length)*(sex + group_size + age + elo + csi) +

(1|population/group/individual/sequence_id)

(Equation 6)

ln(call_duration) ~ ln(sequence_length)*(sex + group_size + age + elo + csi) +

(1|population/group/individual/sequence_id) + (0 + ln(length)|population:group:individual)

The first of these models includes fixed effects for sequence length, sex, group size, age, ELO, and CSI, as well as nested varying intercepts for population, group, individual, and sequence. The second of these adds interactions between the fixed effect for sequence length and sex, group size, age, ELO, and CSI. The third of these adds additional nested varying slopes that allow for the strength of the effect of sequence length on call duration to vary by population, group, and individual. All three of these models were fit to the data, and the one with the lowest AIC is reported in the results.

Importantly, the complex models were only fit to the data from carrion crows. As seen in Table 1, the data from hooded and hybrid crows are too sparse to assess the effect of demographic factors on Menzerath's law. For example, both hooded crows come from the Austria population, and both hybrid crows are female, making it impossible to disentangle those factors from species differences. Analysis code and data are available at https://masonyoungblood.github.io/crow_efficiency/.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Results

In all three subspecies, there is a strong negative relationship between sequence length and call duration, consistent with Menzerath's law (Figure 1 and Table 2). As mentioned in the methods, we chose to not formally analyse species differences in Menzerath's law because of the large disparities in sample size, but it is interesting to note that the strength of the law in hybrid crows is situated between carrion and hooded crows (Figure 1 and Table 2).

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

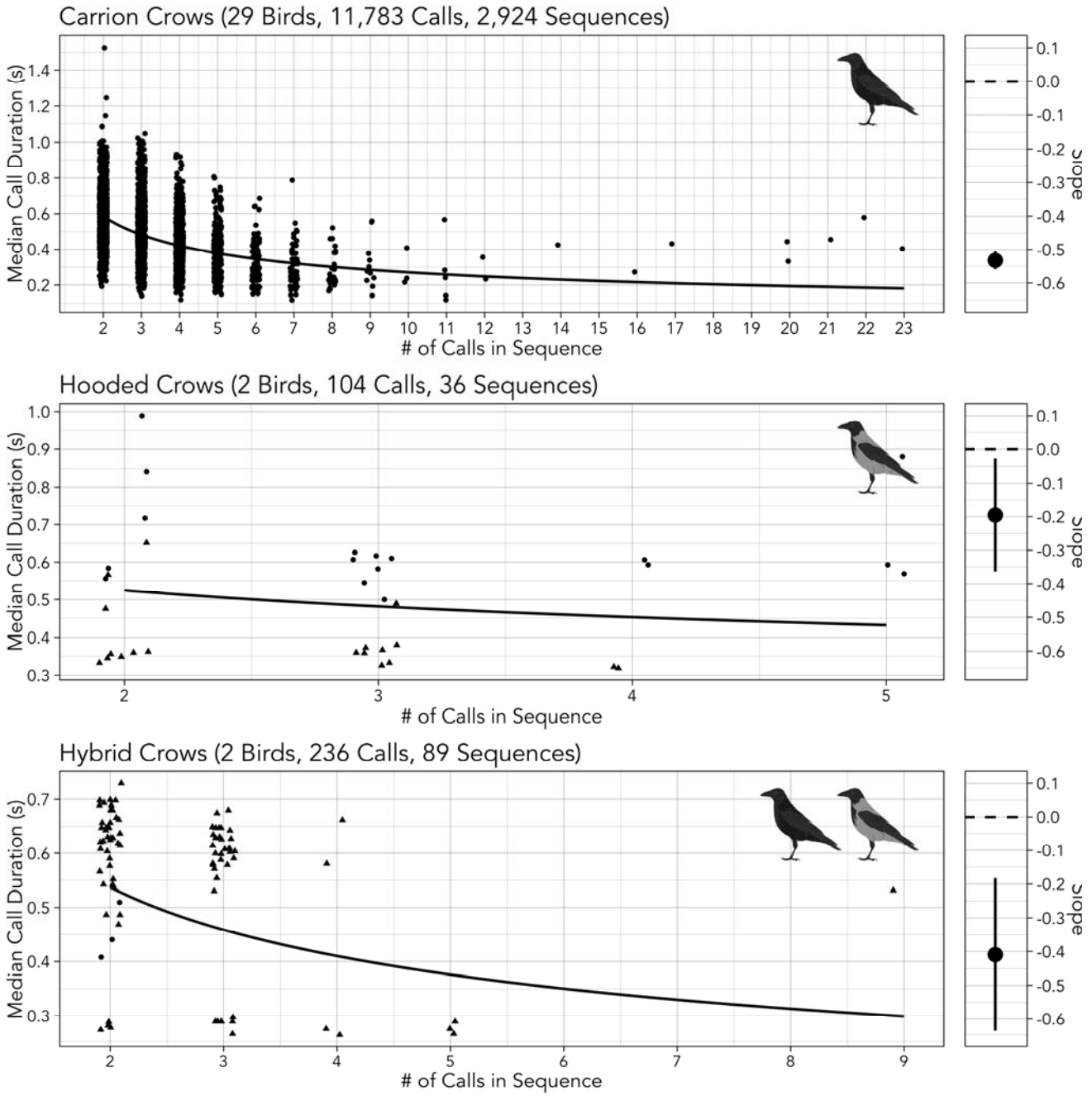


Figure 1: The distribution of median call durations and call sequence lengths (left) and the slope of Menzerath's law (right) for carrion crows (top), hooded crows (middle), and hybrids of carrion and hooded crows (bottom). Different individuals in hooded crows and hybrid crows are plotted in different symbols. The bars in the slope plot (right) mark the 95% confidence intervals around the point estimates.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Table 2: The results of the simple model of Menzerath's law applied separately to call sequences from carrion crows, hooded crows, and hybrids of carrion and hooded crows. 2.5% and 97.5% mark the bounds of the 95% confidence intervals around the point estimates, where intervals that do not overlap zero (i.e., strong effects) are marked with an asterisk.

Species	Parameter	Estimate	2.5% (Lower)	97.5% (Upper)	
Carrion Crows	Intercept	0.018	-0.161	0.197	
	Length	-0.533	-0.559	-0.507	*
Hooded Crows	Intercept	-0.034	-1.609	1.542	
	Length	-0.195	-0.363	-0.027	*
Hybrid Crows	Intercept	-0.218	-0.930	0.494	
	Length	-0.409	-0.637	-0.182	*

Of the three complex models fit to carrion crow calls, the third—with interactions between demographic factors and the effect of sequence length, and with varying slopes for individual identity—fit the data best ($\Delta AIC > 2$). Likelihood ratio tests indicate Equation 6 is the best fitting model of individual variation---Equation 5 outcompetes Equation 4 ($p = 4.5 \times 10^{-32}$), and Equation 6 outcompetes Equation 5 ($p = 1.8 \times 10^{-9}$). The fact that the third complex model (AIC = 14,653) fits better than the second complex model (AIC = 14,698), which excludes varying slopes for individual identity, suggests that there is meaningful individual variation in Menzerath's law, even after accounting for demographic factors. The results of the third complex model indicate that demographic factors influence both call duration and the strength of Menzerath's law (Table 3). Call durations are longer in males, older birds, and more dominant birds, and Menzerath's law appears to be weaker in both males and older birds (Figure 2).

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Table 3: The results of the complex model of Menzerath's law, built to assess individual variation based on demographic factors, applied only to call sequences from carrion crows. 2.5% and 97.5% mark the bounds of the 95% confidence intervals around the point estimates, where intervals that do not overlap zero (i.e., strong effects) are marked with an asterisk. Interactions between demographic factors and Menzerath's law are marked with colons (e.g., "Length : Age" is the interaction between age and effect of sequence length on call duration).

Parameter	Estimate	2.5% (Lower)	97.5% (Upper)	
Intercept	-0.267	-0.494	-0.040	*
Length	-0.678	-0.796	-0.560	*
Sex (M vs. F)	0.541	0.254	0.827	*
Group Size	-0.107	-0.214	0.001	
Age	0.142	0.086	0.198	*
ELO	0.201	0.078	0.324	*
CSI	0.081	-0.087	0.250	
Length : Sex (M vs. F)	0.338	0.183	0.494	*
Length : Group Size	-0.043	-0.108	0.022	
Length : Age	0.066	0.014	0.119	*
Length : ELO	0.045	-0.019	0.109	
Length : CSI	-0.033	-0.114	0.048	

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

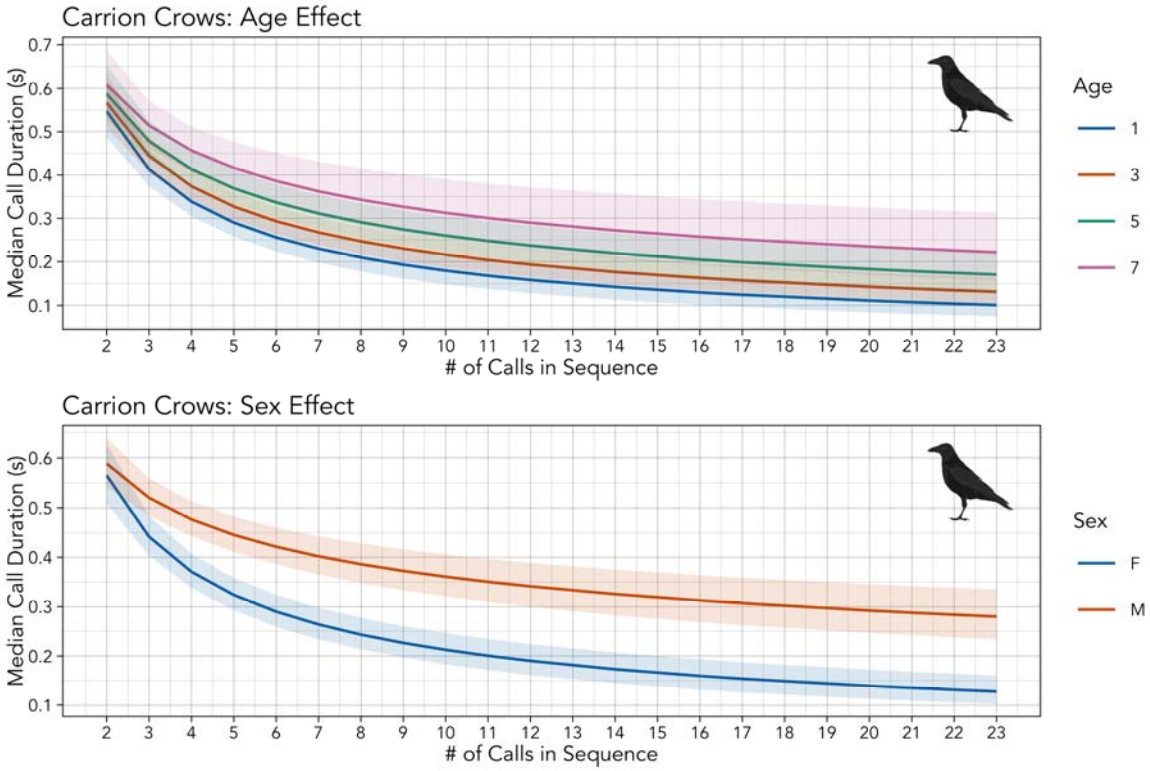


Figure 3: The predicted relationship between element durations and sequence lengths for different levels of age (top) and sex (bottom), from the fitted version of INSERT MODEL.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Discussion

In the present study we show that the call sequences of crows—specifically carrion crows, hooded crows, and hybrids of carrion and hooded crows—adhere to Menzerath's law. In other words, call sequences with more calls are composed of shorter calls, which is suggested to lead to greater communicative efficiency in human language (Piantadosi et al., 2011). To our knowledge, this is the first time efficiency laws have been shown in vocal communication of corvids, adding to the widespread evidence for Menzerath's law in both human and non-human communication systems (Deng et al., 2024; Gustison et al., 2016; Semple et al., 2010; Youngblood, 2024, 2024, 2025). Interestingly, within carrion crows, adherence to Menzerath's Law was modulated by sex and age. Males' calls are longer, and their adherence to Menzerath's law is weaker, which may reflect sex-specific roles in social organization or differences in vocal usage patterns. Linguistic laws previously mostly have been investigated in song, which in many species is sexually selected in males to attract females (Clink & Lau, 2020; James et al., 2021; Valente et al., 2021; Youngblood, 2024, 2025). Calls in corvids often function as contact calls (Anjos & Vielliard, 1993; Berger & Ligon, 1977; McCaig et al., 2015; Roskaft & Espmark, 1982; Sitasuwan & Thaler, 1985), food calls (Bugnyar et al., 2001; Mates et al., 2015; Roehmholdt, 2019; Roskaft & Espmark, 1982; Szipl & Bugnyar, 2014), territory calls (Mates et al., 2015; Sabol et al., 2022; Tanimoto et al., 2017), or alert signals (Chamberlain & Cornwell, 1971; Conner, 1985; Cornell et al., 2012; Ellis, 2009; Mates et al., 2015; Pendergraft & Marzluff, 2019; Tanimoto et al., 2017; Yorzinski & Vehrencamp, 2009) and are typically emitted by both sexes. In rooks, males produced call units with lower diversity and gradation than females and individual males produced different call repertoires, whereas females produced more similar call repertoires to each other (Martin et al., 2024). Our study indicates that female carrion crows might communicate more efficiently compared to males. This is different to a previous study in lemurs, *Indri indri*, which showed no sex difference in adherence to Menzerath's law (Valente et al., 2021).

Surprisingly, younger individuals displayed stronger adherence to Menzerath's Law compared to older crows. This finding contrasts with our original prediction, that efficiency in vocal production would

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

366 increase with age. This finding contrasts previous findings in zebra finches, *Taeniopygia castanotis*,
367 were song elements and gaps shorten over development (Glaze & Troyer, 2013) and humans, were
368 vocal efficiency increases with age (Tang & Stathopoulos, 1995). Songbirds reared without auditory
369 experiences that guide vocal development still adhere to Menzerath's law, indicating that vocal
370 efficiency is linked to vocal production mechanisms rather than learned behaviour (James et al., 2021).
371 Our results could indicate that carrion crow vocalisations are limited to vocal efficiency at younger age
372 and become more flexible with increased age. Since call duration overall also increases with age in our
373 analysis, we suggest the need for informativeness of calls may outweigh the importance of efficiency as
374 crows age. Further studies on the ontogeny of adherence to Menzerath's law are important.
375
376 Group size, dominance status, and the strength of affiliative relationships did not significantly influence
377 adherence to Menzerath's Law, although call duration itself was longer in more dominant birds. This
378 suggests that, in carrion crows, the structural efficiency of vocal sequences may be relatively stable
379 across different social contexts. However, it should be noted that our data stems from captive
380 individuals, where social context such as group composition were under human control, hence the full
381 range of social contexts might not be captured in the data. Overall, our findings highlight the presence
382 of Menzerath's Law in carrion crow communication and point to individual-level social factors, rather
383 than broader group dynamics, as influencing the degree of adherence. Future research should explore
384 whether these patterns hold across different communicative contexts such as different call-types and
385 in other corvid species, to better understand the interplay between social structure, cognitive
386 demands, and communicative efficiency.
387
388
389
390
391
392

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

References

Altmann, G. (1980). Prolegomena to Menzerath's law. *Glottometrika*, *2*(2), 1–10.

Anjos, L. D., & Vielliard, J. M. E. (1993). Repertoire of the acoustic communication of the azure jay Cyanocorax caeruleus (Vieillot) (Aves, Corvidae). *Revista Brasileira de Zoologia*, *10*(4), 657–664. https://doi.org/10.1590/S0101-81751993000400011

Archie, E. A., Tung, J., Clark, M., Altmann, J., & Alberts, S. C. (2014). Social affiliation matters: Both same-sex and opposite-sex relationships predict survival in wild female baboons. *Proceedings of the Royal Society B: Biological Sciences*, *281*(1793), 20141261. https://doi.org/10.1098/rspb.2014.1261

AviList Core Team. (2025). *AviList: The Global Avian Checklist, v2025*. [Dataset]. https://doi.org/10.2173/avilist.v2025

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, *67*(1), 1–48.

Berger, L. R., & Ligon, J. D. (1977). Vocal communication and individual recognition in the pinon jay, Gymnorhinus cyanocephalus. *Animal Behaviour*, *25*, 567–584. https://doi.org/10.1016/0003-3472(77)90107-5

Bezerra, B. M., Souto, A. S., Radford, A. N., & Jones, G. (2011). Brevity is not always a virtue in primate communication. *Biology Letters*, *7*(1), 23–25. https://doi.org/10.1098/rsbl.2010.0455

Brecht, K. F., Hage, S. R., Gavrilov, N., & Nieder, A. (2019). Volitional control of vocalizations in corvid songbirds. *PLOS Biology*, *17*(8), e3000375. https://doi.org/10.1371/journal.pbio.3000375

Brenowitz, E. A., Margoliash, D., & Nordeen, K. W. (1997). An introduction to birdsong and the avian song system. *Journal of Neurobiology*, *33*(5), 495–500. https://doi.org/10.1002/(SICI)1097-4695(19971105)33:5<495::AID-NEU1>3.0.CO;2-#

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

418 Bugnyar, T., Kijne, M., & Kotrschal, K. (2001). Food calling in ravens: Are yells referential
419 signals? *Animal Behaviour*, *61*(5), 949–958. https://doi.org/10.1006/anbe.2000.1668
420 Chamberlain, D. R., & Cornwell, G. W. (1971). Selected vocalizations of the common crow.
421 *The Auk*, *88*, 613–634.
422 Choi, N., Adams, M., Fowler-Finn, K., Knowlton, E., Rosenthal, M., Rundus, A., Santer, R. D.,
423 Wilgers, D., & Hebets, E. A. (2022). Increased signal complexity is associated with
424 increased mating success. *Biology Letters*, *18*(5), 20220052.
425 https://doi.org/10.1098/rsbl.2022.0052
426 Clink, D. J., & Lau, A. R. (2020). Adherence to Menzerath's Law is the exception (not the rule)
427 in three duetting primate species. *Royal Society Open Science*, *7*(11), 201557.
428 https://doi.org/10.1098/rsos.201557
429 Conner, R. N. (1985). Vocalizations of common ravens in Virginia. *The Condor*, *87*(3), 379–
430 388. https://doi.org/10.2307/1367219
431 Cornell, H. N., Marzluff, J. M., & Pecoraro, S. (2012). Social learning spreads knowledge about
432 dangerous humans among American crows. *Proceedings of the Royal Society B:*
433 *Biological Sciences*, *279*(1728), 499–508. https://doi.org/10.1098/rspb.2011.0957
434 Demartsev, V., Gordon, N., Barocas, A., Bar-Ziv, E., Ilany, T., Goll, Y., Ilany, A., & Geffen, E.
435 (2019). The "Law of Brevity" in animal communication: Sex-specific signaling
436 optimization is determined by call amplitude rather than duration. *Evolution Letters*,
437 *3*(6), 623–634. https://doi.org/10.1002/evl3.147
438 Deng, K., He, Y.-X., Wang, X.-P., Wang, T.-L., Wang, J.-C., Chen, Y.-H., & Cui, J.-G. (2024).
439 Hainan frilled treefrogs' calls partially conform to Menzerath–Altmann's law, but
440 oppose Zipf's law of abbreviation. *Animal Behaviour*, *213*, 51–59.
441 https://doi.org/10.1016/j.anbehav.2024.04.011

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Ellis, J. M. S. (2009). Anti□predator signals as advertisements: Evidence in white□throated magpie□jays. *Ethology*, *115*(6), 522–532. https://doi.org/10.1111/j.1439-0310.2009.01631.x

Endler, J. A. (1993). Some general comments on the evolution and design of animal communication systems. *Philosophical Transactions: Biological Sciences*, *340*(1292), 215–225.

Favaro, L., Gamba, M., Cresta, E., Fumagalli, E., Bandoli, F., Pilenga, C., Isaja, V., Mathevon, N., & Reby, D. (2020). Do penguins' vocal sequences conform to linguistic laws? *Biology Letters*, *16*(2), 20190589. https://doi.org/10.1098/rsbl.2019.0589

Ferrer□i□Cancho, R., Hernández□Fernández, A., Lusseau, D., Agoramoorthy, G., Hsu, M. J., & Semple, S. (2013). Compression as a universal principle of animal behavior. *Cognitive Science*, *37*(8), 1565–1578. https://doi.org/10.1111/cogs.12061

Freeberg, T. M., & Krams, I. (2015). Does social complexity link vocal complexity and cooperation? *Journal of Ornithology*, *156*(S1), 125–132. https://doi.org/10.1007/s10336-015-1233-2

G. Torre, I., Dębowski, Ł., & Hernández-Fernández, A. (2021). Can Menzerath's law be a criterion of complexity in communication? *PLOS ONE*, *16*(8), e0256133. https://doi.org/10.1371/journal.pone.0256133

Gibson, E., Futrell, R., Piantadosi, S. P., Dautriche, I., Mahowald, K., Bergen, L., & Levy, R. (2019). How efficiency shapes human language. *Trends in Cognitive Sciences*, *23*(5), 389–407. https://doi.org/10.1016/j.tics.2019.02.003

Glaze, C. M., & Troyer, T. W. (2013). Development of temporal structure in zebra finch song. *Journal of Neurophysiology*, *109*(4), 1025–1035. https://doi.org/10.1152/jn.00578.2012

Gustison, M. L., Semple, S., Ferrer-i-Cancho, R., & Bergman, T. J. (2016). Gelada vocal sequences follow Menzerath's linguistic law. *Proceedings of the National Academy of Sciences*, *113*(19). https://doi.org/10.1073/pnas.1522072113

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

468 Heesen, R., Hobaiter, C., Ferrer-i-Cancho, R., & Semple, S. (2019). Linguistic laws in
469 chimpanzee gestural communication. *Proceedings of the Royal Society B: Biological*
470 *Sciences*, *286*(1896), 20182900. https://doi.org/10.1098/rspb.2018.2900
471 Hernández-Fernández, A., G. Torre, I., Garrido, J.-M., & Lacasa, L. (2019). Linguistic Laws in
472 speech: The case of Catalan and Spanish. *Entropy*, *21*(12), 1153.
473 https://doi.org/10.3390/e21121153
474 Hou, R., Huang, C.-R., Do, H. S., & Liu, H. (2017). A study on correlation between Chinese
475 sentence and constituting clauses based on the Menzerath-Altmann law. *Journal of*
476 *Quantitative Linguistics*, *24*(4), 350–366.
477 https://doi.org/10.1080/09296174.2017.1314411
478 Huang, M., Ma, H., Ma, C., Garber, P. A., & Fan, P. (2020). Male gibbon loud morning calls
479 conform to Zipf's law of brevity and Menzerath's law: Insights into the origin of human
480 language. *Animal Behaviour*, *160*, 145–155.
481 https://doi.org/10.1016/j.anbehav.2019.11.017
482 Igamberdiev, A. U. (2023). Overcoming the limits of natural computation in biological
483 evolution toward the maximization of system efficiency. *Biological Journal of the*
484 *Linnean Society*, *139*(4), 539–554. https://doi.org/10.1093/biolinnean/blac093
485 James, L. S., Mori, C., Wada, K., & Sakata, J. T. (2021). Phylogeny and mechanisms of shared
486 hierarchical patterns in birdsong. *Current Biology*, *31*(13), 2796-2808.e9.
487 https://doi.org/10.1016/j.cub.2021.04.015
488 K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology. (2024).
489 *Raven Pro: Interactive Sound Analysis Software [Computer software]*. (Version 1.6.5)
490 [Computer software]. https://www.ravensoundsoftware.com/
491 Krams, I., Krama, T., Freeberg, T. M., Kullberg, C., & Lucas, J. R. (2012). Linking social
492 complexity and vocal complexity: A parid perspective. *Philosophical Transactions of*

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

493 *the Royal Society B: Biological Sciences*, *367*(1597), 1879–1891.
494 https://doi.org/10.1098/rstb.2011.0222
495 Lewis, R. N., Kwong, A., Soma, M., De Kort, S. R., & Gilman, R. T. (2023). *Java sparrow*
496 *song conforms to Menzerath's Law but not Zipf's Law of Abbreviation*.
497 https://doi.org/10.1101/2023.12.13.571437
498 Liao, D. A., Brecht, K. F., Veit, L., & Nieder, A. (2024). Crows "count" the number of self-
499 generated vocalizations. *Science*, *384*(6698), 874–877.
500 https://doi.org/10.1126/science.adl0984
501 Luef, E. M., Ter Maat, A., & Pika, S. (2017). Vocal similarity in long-distance and short-
502 distance vocalizations in raven pairs (Corvus corax) in captivity. *Behavioural*
503 *Processes*, *142*, 1–7. https://doi.org/10.1016/j.beproc.2017.05.013
504 Manser, M. B., Jansen, D. A. W. A. M., Graw, B., Hollén, L. I., Bousquet, C. A. H., Furrer, R.
505 D., & Le Roux, A. (2014). Vocal Complexity in Meerkats and Other Mongoose Species.
506 In *Advances in the Study of Behavior* (Vol. 46, pp. 281–310). Elsevier.
507 https://doi.org/10.1016/B978-0-12-800286-5.00006-7
508 Martin, K., Cornero, F. M., Clayton, N. S., Adam, O., Obin, N., & Dufour, V. (2024). Vocal
509 complexity in a socially complex corvid: Gradation, diversity and lack of common call
510 repertoire in male rooks. *Royal Society Open Science*, *11*(1), 231713.
511 https://doi.org/10.1098/rsos.231713
512 Mates, E. A., Tarter, R. R., Ha, J. C., Clark, A. B., & McGowan, K. J. (2015). Acoustic
513 profiling in a complexly social species, the American crow: Caws encode information
514 on caller sex, identity and behavioural context. *Bioacoustics*, *24*(1), 63–80.
515 https://doi.org/10.1080/09524622.2014.933446
516 McCaig, T., Brown, M., & Jones, D. N. (2015). Exploring possible functions of vocalisations in
517 the Torresian Crow Corvus orru. *Australian Field Ornithology*, *32*, 201–208.
518 Menzerath, P. (1954). *Die Architektonik des deutschen Wortschatzes*. Dümmler.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

519 Milička, J. (2023). Menzerath's law: Is it just regression toward the mean? *Glottometrics*, *55*, 1–
520 16. https://doi.org/10.53482/2023_55_409
521 Neumann, C., Duboscq, J., Dubuc, C., Ginting, A., Irwan, A. M., Agil, M., Widdig, A., &
522 Engelhardt, A. (2011). Assessing dominance hierarchies: Validation and advantages of
523 progressive evaluation with Elo-rating. *Animal Behaviour*, *82*(4), 911–921.
524 https://doi.org/10.1016/j.anbehav.2011.07.016
525 Peckre, L., Kappeler, P. M., & Fichtel, C. (2019). Clarifying and expanding the social
526 complexity hypothesis for communicative complexity. *Behavioral Ecology and*
527 *Sociobiology*, *73*(1), 11. https://doi.org/10.1007/s00265-018-2605-4
528 Pendergraft, L. T., & Marzluff, J. M. (2019). Fussing over food: Factors affecting the
529 vocalizations American crows utter around food. *Animal Behaviour*, *150*, 39–57.
530 https://doi.org/10.1016/j.anbehav.2019.01.024
531 Piantadosi, S. T., Tily, H., & Gibson, E. (2011). Word lengths are optimized for efficient
532 communication. *Proceedings of the National Academy of Sciences*, *108*(9), 3526–3529.
533 https://doi.org/10.1073/pnas.1012551108
534 Pollard, K. A., & Blumstein, D. T. (2012). Evolving communicative complexity: Insights from
535 rodents and beyond. *Philosophical Transactions of the Royal Society B: Biological*
536 *Sciences*, *367*(1597), 1869–1878. https://doi.org/10.1098/rstb.2011.0221
537 R Core Team. (2019). *R: a language and environment for statistical computing.* (Version 4.3.1.)
538 [Computer software]. http://www.r723 project.org/
539 Risueno-Segovia, C., Dohmen, D., Gultekin, Y. B., Pomberger, T., & Hage, S. R. (2023).
540 Linguistic law-like compression strategies emerge to maximize coding efficiency in
541 marmoset vocal communication. *Proceedings of the Royal Society B: Biological*
542 *Sciences*, *290*(2007), 20231503. https://doi.org/10.1098/rspb.2023.1503
543 Roehmholdt, C. E. (2019). *Using vocalizations to determine the capacity for future cognition in*
544 *the common raven (Corvus Corax)* [Doctoral Thesis]. Evergreen State College.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

545 Roskaft, E., & Espmark, Y. (1982). Vocal communication by the rook Corvus frugilegus during
546 the breeding season. *Ornis Scandinavica*, *13*(1), 38. https://doi.org/10.2307/3675971
547 Sabol, A. C., Greggor, A. L., Masuda, B., & Swaisgood, R. R. (2022). Testing the maintenance
548 of natural responses to survival-relevant calls in the conservation breeding population of
549 a critically endangered corvid (Corvus hawaiiensis). *Behavioral Ecology and*
550 *Sociobiology*, *76*(1), 21. https://doi.org/10.1007/s00265-022-03130-8
551 Safryghin, A., Cross, C., Fallon, B., Heesen, R., Ferrer-i-Cancho, R., & Hobaiter, C. (2022).
552 Variable expression of linguistic laws in ape gesture: A case study from chimpanzee
553 sexual solicitation. *Royal Society Open Science*, *9*(11), 220849.
554 https://doi.org/10.1098/rsos.220849
555 Sánchez-Tójar, A., Schroeder, J., & Farine, D. R. (2018). A practical guide for inferring reliable
556 dominance hierarchies and estimating their uncertainty. *Journal of Animal Ecology*,
557 *87*(3), 594–608. https://doi.org/10.1111/1365-2656.12776
558 Sandoval, L., & Graham, B. (2025). Songs and calls: Perspectives on creating a global
559 definition. *Ornitología Neotropical*, *35*(2). https://doi.org/10.58843/ornneo.v35i2.1361
560 Semple, S., Ferrer-i-Cancho, R., & Gustison, M. L. (2022). Linguistic laws in biology. *Trends*
561 *in Ecology & Evolution*, *37*(1), 53–66. https://doi.org/10.1016/j.tree.2021.08.012
562 Semple, S., Hsu, M. J., & Agoramoorthy, G. (2010). Efficiency of coding in macaque vocal
563 communication. *Biology Letters*, *6*(4), 469–471. https://doi.org/10.1098/rsbl.2009.1062
564 Silk, J. B., Beehner, J. C., Bergman, T. J., Crockford, C., Engh, A. L., Moscovice, L. R., Wittig,
565 R. M., Seyfarth, R. M., & Cheney, D. L. (2010). Female chacma baboons form strong,
566 equitable, and enduring social bonds. *Behavioral Ecology and Sociobiology*, *64*(11),
567 1733–1747. https://doi.org/10.1007/s00265-010-0986-0
568 Sitasuwan, N., & Thaler, E. (1985). Vocal inventory and communication in the Chough
569 (Pyrrhocorax pyrrhocorax), in the Alpine Chough (P. graculus) and in their hybrids.
570 *Journal für Ornithologie*, *126*, 181–193.

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

571 Stepanov, A., Zhivomirov, H., Nedelchev, I., & Stateva, P. (2023). *Bottlenose dolphins'*
572 *broadband clicks are structured for communication*.
573 https://doi.org/10.1101/2023.01.11.523588
574 Suzuki, T. N. (2014). Communication about predator type by a bird using discrete, graded and
575 combinatorial variation in alarm calls. *Animal Behaviour*, *87*, 59–65.
576 https://doi.org/10.1016/j.anbehav.2013.10.009
577 Szipl, G., & Bugnyar, T. (2014). Craving ravens: Individual 'haa' call rates at feeding sites as
578 cues to personality and levels of fission-fusion dynamics? *Animal Behavior and*
579 *Cognition*, *1*(3), 265. https://doi.org/10.12966/abc.08.04.2014
580 Szipl, G., Ringler, E., & Bugnyar, T. (2018). Attacked ravens flexibly adjust signalling
581 behaviour according to audience composition. *Proceedings of the Royal Society B:*
582 *Biological Sciences*, *285*(1880), 20180375. https://doi.org/10.1098/rspb.2018.0375
583 Tang, J., & Stathopoulos, E. T. (1995). Vocal efficiency as a function of vocal intensity: A
584 study of children, women, and men. *The Journal of the Acoustical Society of America*,
585 *97*(3), 1885–1892. https://doi.org/10.1121/1.412062
586 Tanimoto, A. M., Hart, P. J., Pack, A. A., & Switzer, R. (2017). Vocal repertoire and signal
587 characteristics of 'Alalā, the Hawaiian Crow (*Corvus hawaiiensis*). *The Wilson Journal*
588 *of Ornithology*, *129*(1), 25–35. https://doi.org/10.1676/1559-4491-129.1.25
589 Torre, I. G., Luque, B., Lacasa, L., Kello, C. T., & Hernández-Fernández, A. (2019). On the
590 physical origin of linguistic laws and lognormality in speech. *Royal Society Open*
591 *Science*, *6*(8), 191023. https://doi.org/10.1098/rsos.191023
592 Valente, D., De Gregorio, C., Favaro, L., Friard, O., Miaretsoa, L., Raimondi, T., Ratsimbazafy,
593 J., Torti, V., Zanoli, A., Giacoma, C., & Gamba, M. (2021). Linguistic laws of brevity:
594 Conformity in Indri indri. *Animal Cognition*, *24*(4), 897–906.
595 https://doi.org/10.1007/s10071-021-01495-3

bioRxiv preprint doi: https://doi.org/10.1101/2025.07.03.663010; this version posted July 3, 2025. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

596 Vradi, A. A. (2021). *Dolphin communication: A quantitative linguistics approach* [Master's
597 thesis]. Universitat Politècnica de Catalunya.
598 Wascher, C. A. F. (2021). Association between social factors and gastrointestinal parasite
599 product excretion in a group of non-cooperatively breeding carrion crows. *Behavioral*
600 *Ecology and Sociobiology*, *75*(2), 30. https://doi.org/10.1007/s00265-021-02967-9
601 Wascher, C. A. F., Canestrari, D., & Baglione, V. (2019). Affiliative social relationships and
602 coccidian oocyst excretion in a cooperatively breeding bird species. *Animal Behaviour*,
603 *158*, 121–130. https://doi.org/10.1016/j.anbehav.2019.10.009
604 Wascher, C. A. F., & Reynolds, S. (2025). Vocal communication in corvids: A systematic
605 review. *Animal Behaviour*, *221*, 123073. https://doi.org/10.1016/j.anbehav.2024.123073
606 Wascher, C. A. F., Waterhouse, G., & Beheim, B. A. (2025). Vocal mimicry in corvids.
607 *BioRxiv*. https://doi.org/10.1101/2025.03.26.645457
608 Yorzinski, J. L., & Vehrencamp, S. L. (2009). The effect of predator type and danger level on
609 the mob calls of the American crow. *The Condor*, *111*(1), 159–168.
610 https://doi.org/10.1525/cond.2009.080057
611 Youngblood, M. (2024). Language-like efficiency and structure in house finch song.
612 *Proceedings of the Royal Society B: Biological Sciences*, *291*(2020), 20240250.
613 https://doi.org/10.1098/rspb.2024.0250
614 Youngblood, M. (2025). Language-like efficiency in whale communication. *Science Advances*,
615 *11*, eads6014. https://doi.org/10.31234/osf.io/tduab
616 Zhang, C., Zheng, Z., Lucas, J. R., Wang, Y., Fan, X., Zhao, X., Feng, J., Sun, C., & Jiang, T.
617 (2024). Do bats' social vocalizations conform to Zipf's law and the Menzerath-Altmann
618 law? *iScience*, *27*(7), 110401. https://doi.org/10.1016/j.isci.2024.110401
619 Zipf, G. K. (1949). *Human behavior and the principle of least effort: An introduction to human*
620 *ecology*. Addison-Wesley Press.
621